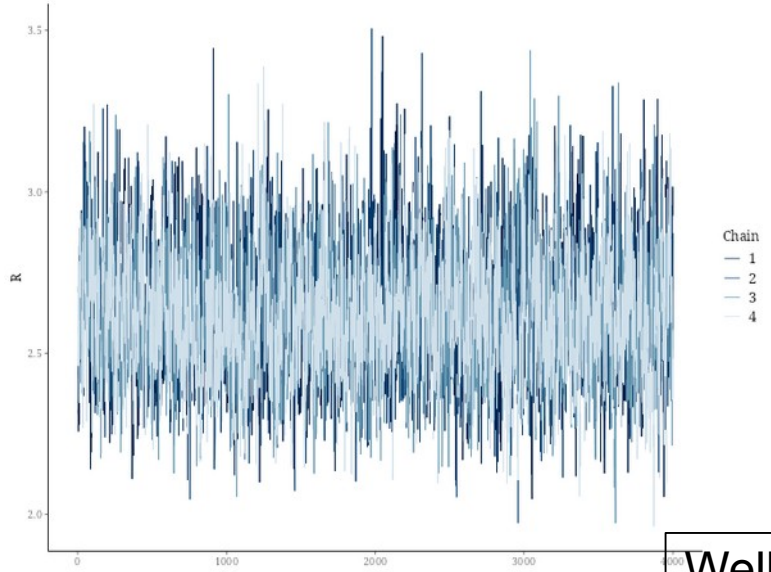
Today

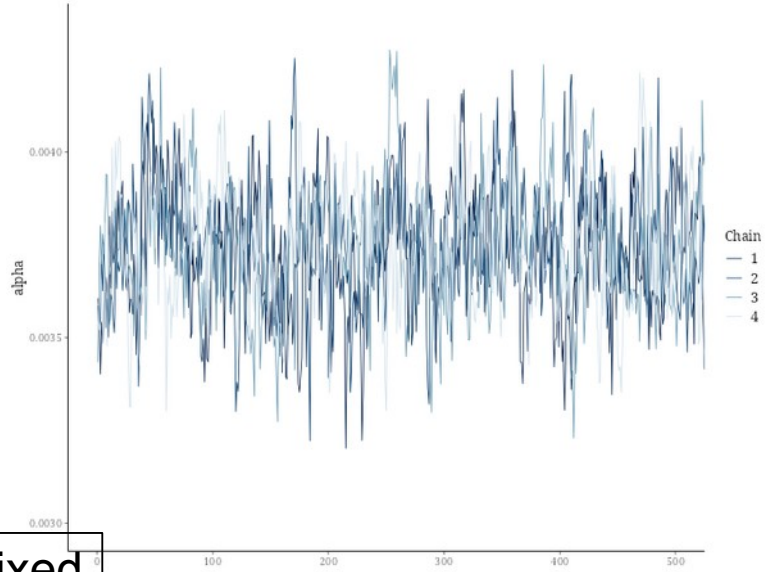
- Model with complex stochasticity
- Training with Rosenbluth in JAGS
- Training with HMC in Stan
- Model checks
- Model comparison
  - LOOCV
  - Bayesian LOOIC

Train NBBG with HMC (Stan)

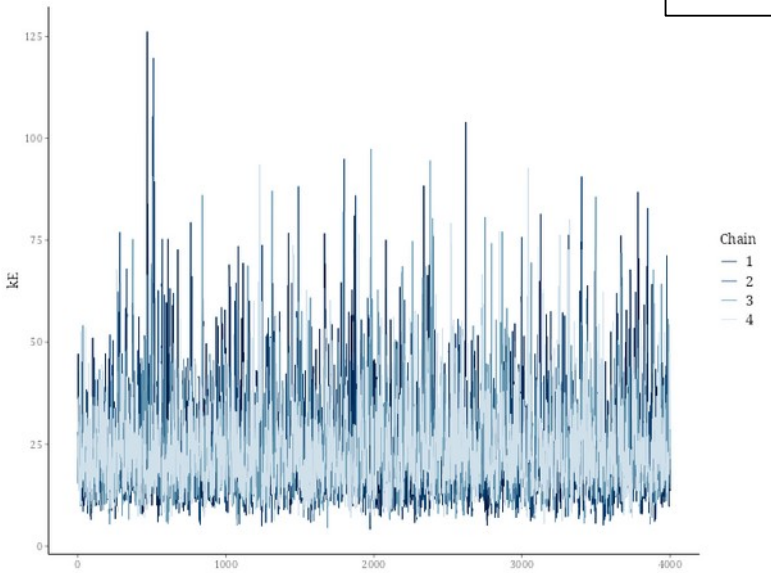
```
> summarise_draws(samples_array, mean, sd, rhat, ess_bulk, ess_tail, mcse_mean)
# A tibble: 4 × 7
  variable     mean        sd  rhat ess_bulk ess_tail  mcse_mean
  <chr>       <dbl>     <dbl> <dbl>    <dbl>    <dbl>      <dbl>
1 R         2.62     0.187     1.00    2584.    5438. 0.00366
2 alpha     0.00373  0.000163  1.00    2730.    5311. 0.00000312
3 kD        3.12    20.3       1.00    7653.    6363. 0.298
4 kE       23.8     11.1       1.00    2608.    4941. 0.216
```

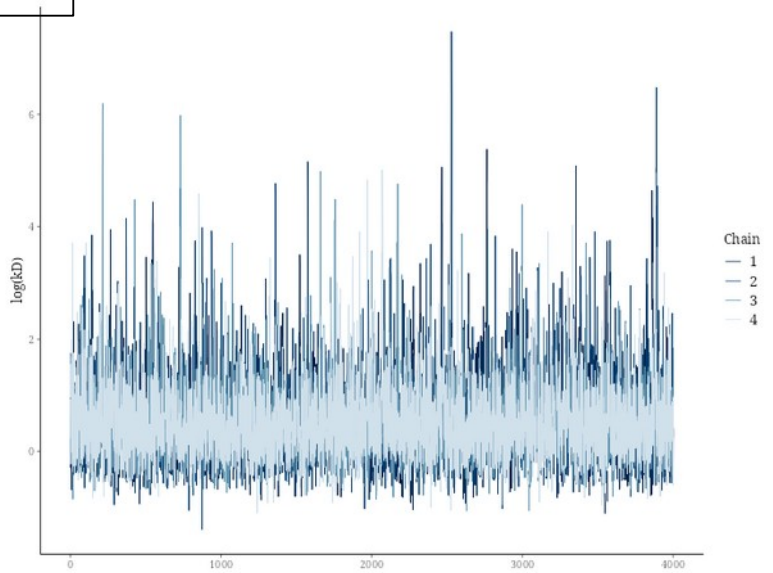
Train NBBG with HMC (Stan)



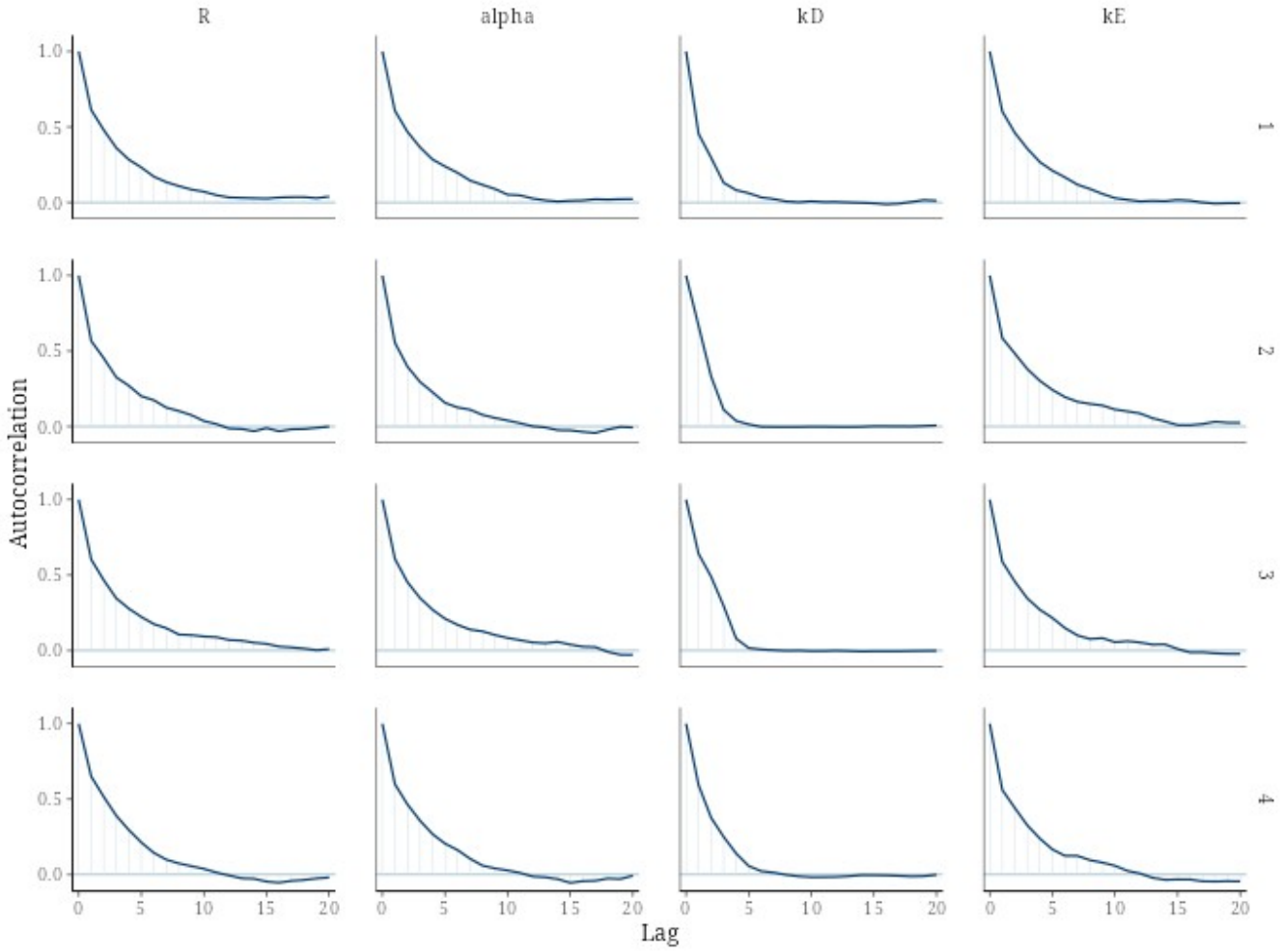


Well mixed

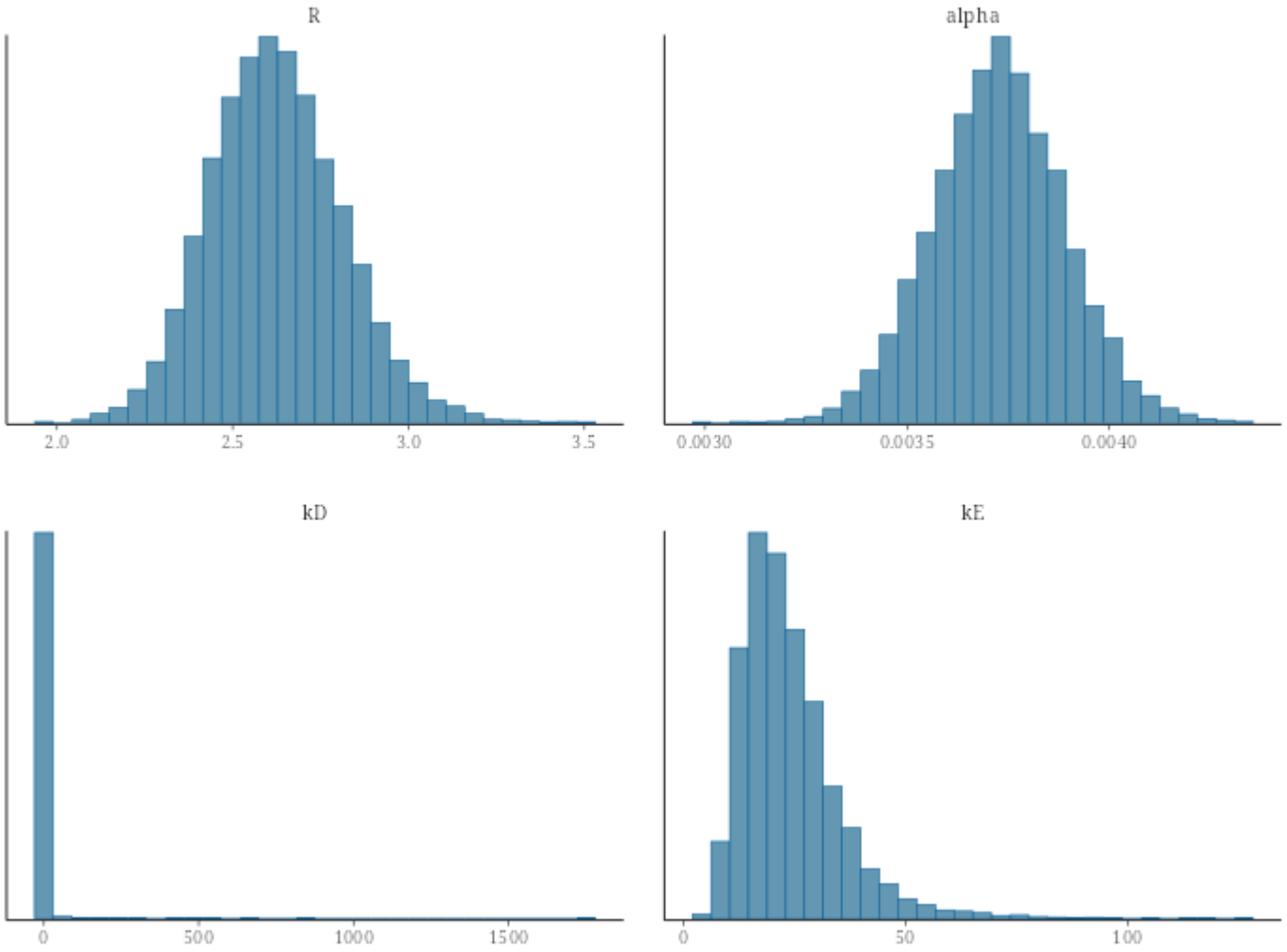




Train NBBG with HMC (Stan)



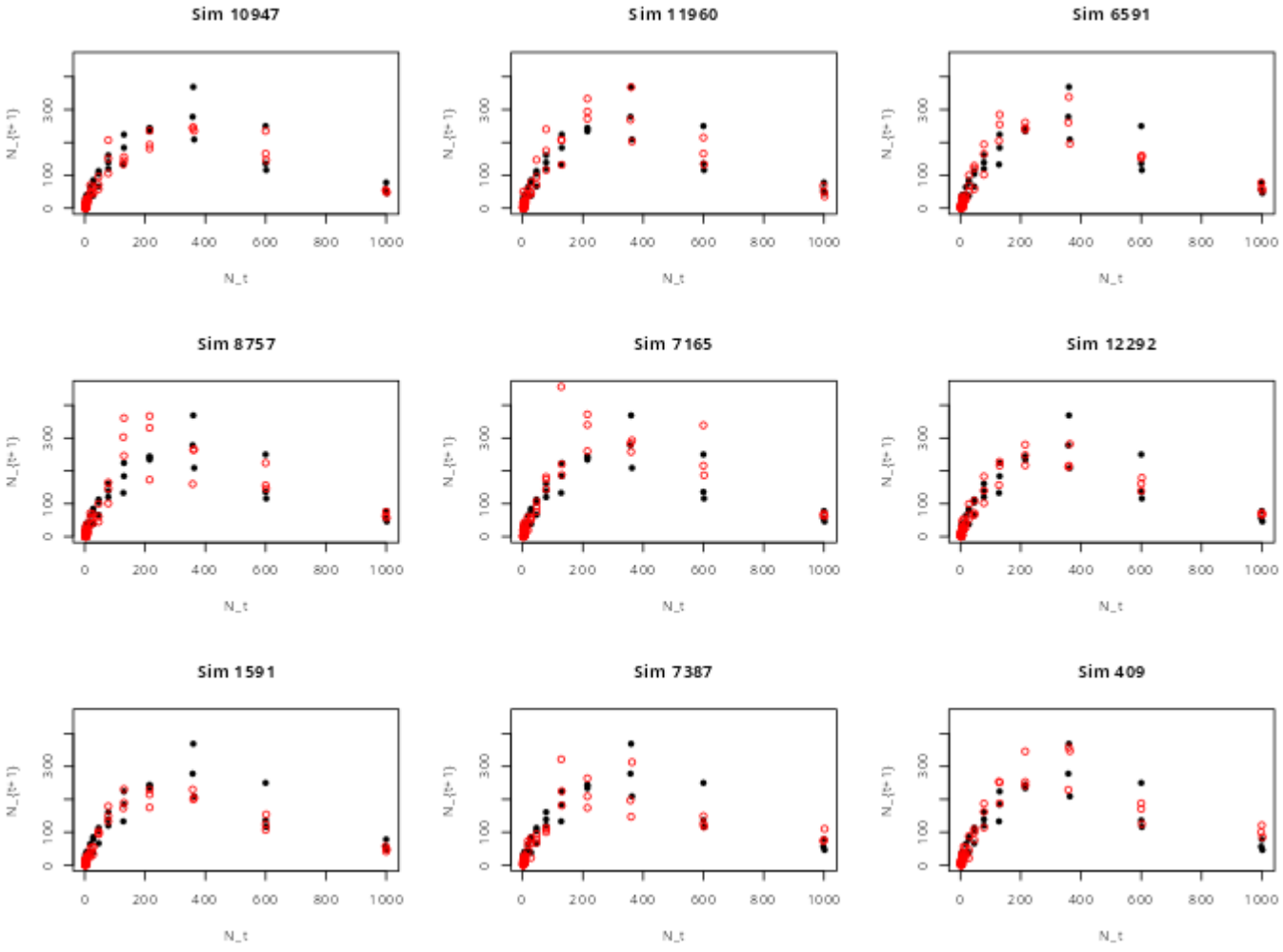
Posterior NBBG



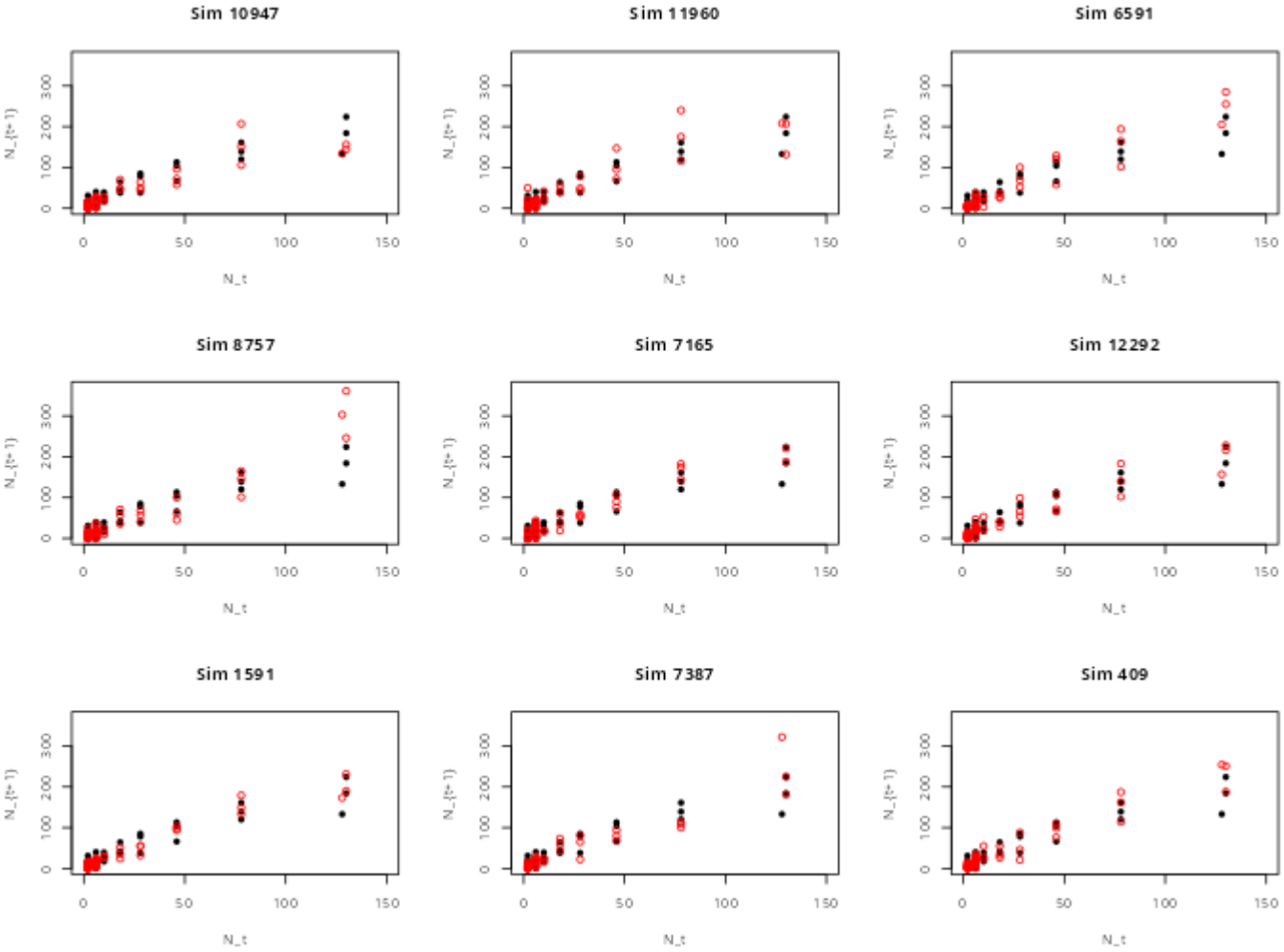
Posterior NBBG

```
> quantile(samples$R, c(0.025, 0.975))
    2.5%    97.5%
2.280279 3.012783
> quantile(samples$alpha, c(0.025, 0.975))
       2.5%       97.5%
0.003407321 0.004050239
> HPDinterval(as.mcmc(samples$kD))
         lower    upper
var1 0.3496306 6.289069
attr(,"Probability")
[1] 0.95
> HPDinterval(as.mcmc(samples$kE))
        lower    upper
var1 7.192827 44.95648
attr(,"Probability")
[1] 0.95
```

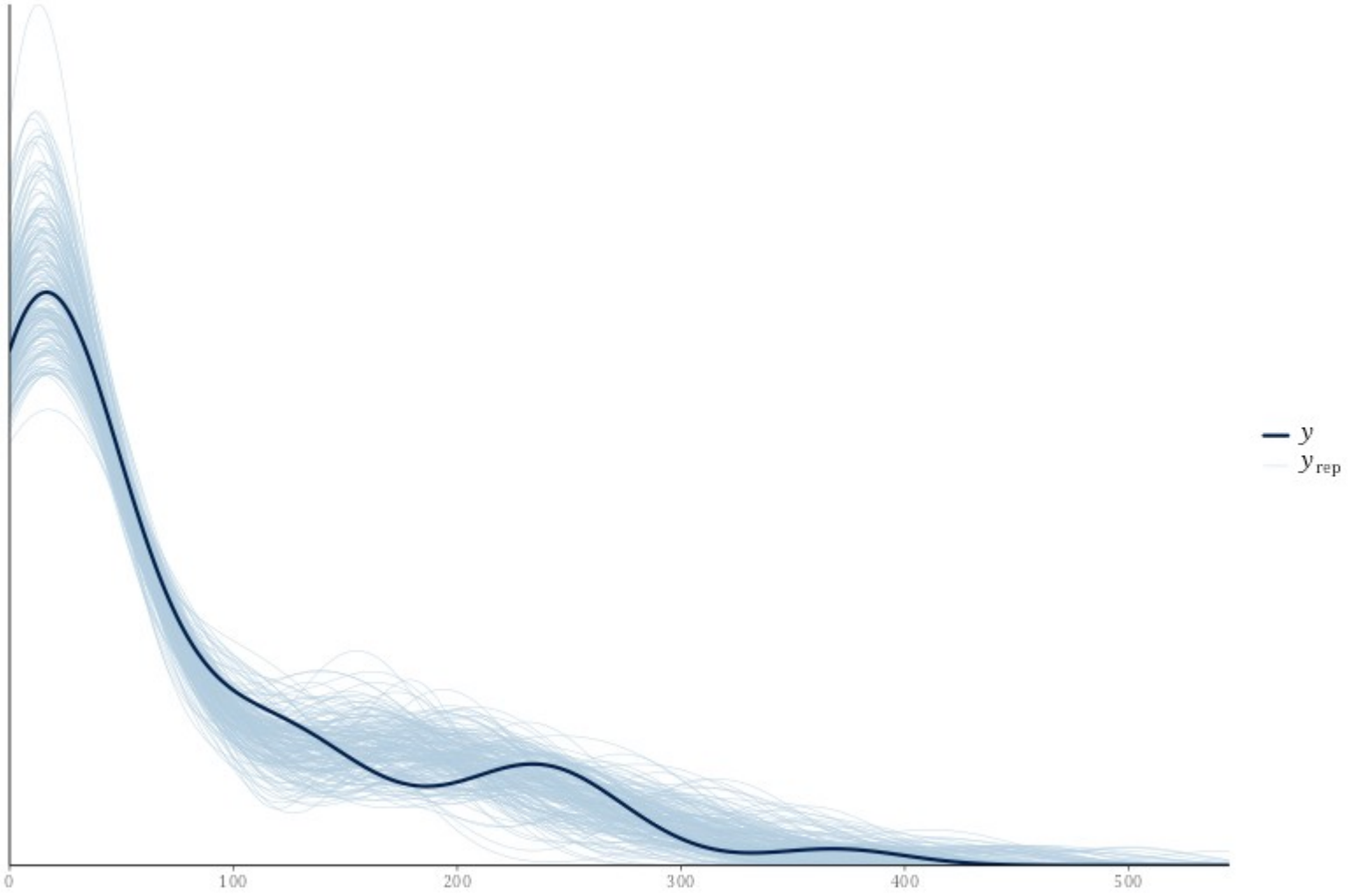
NBBG model checks



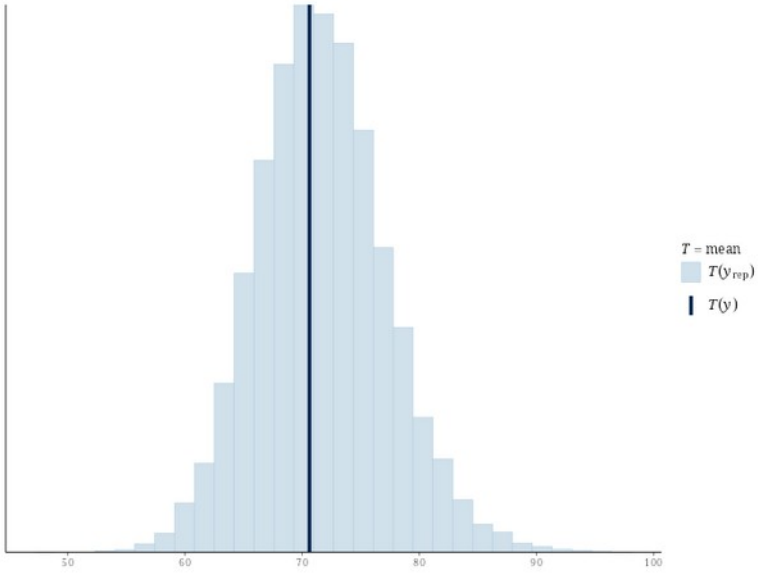
NBBG model checks

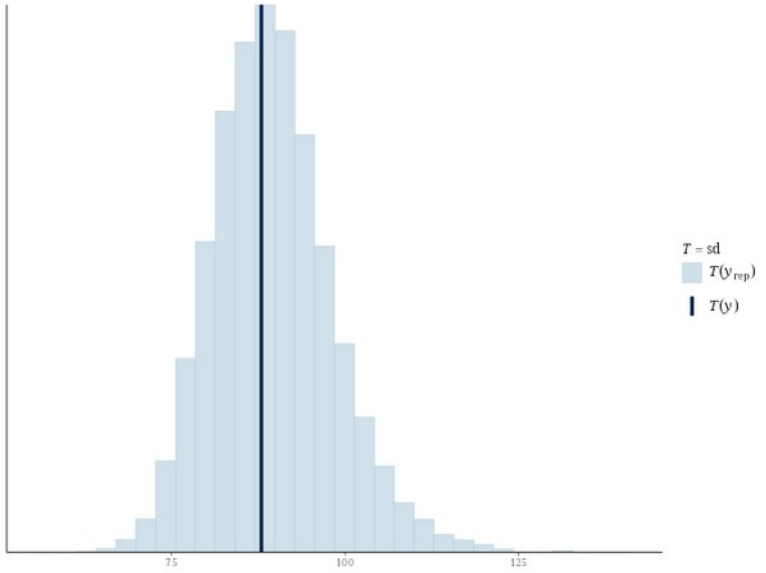


NBBG model checks



NBBG model checks





Bayesian LOOIC

Leave-one-out information criterion: LOOIC = -2 $elpd_{loo}$

elpd: expected log predictive density, out of sample
"density": pdf (density) or pmf (probability)
loo: leave one out is one way to be out of sample

$$\text{elpd}_{\text{loo}} = \sum_{i=1}^{n} \log \left(\frac{1}{S} \sum_{s=1}^{S} p(y_i|\theta_{-i}^{(s)}) \right)$$

Sum is expectation for dataset

Predictive density
Monte Carlo estimate (mean of samples)
s: sample from the posterior

Predictive performance:

$$p(y_i|\theta_{-i}^{(s)})$$

Probability or density of the held out data point y_i, given the sample s of parameters, θ, for the model trained without data point i.
(similar to a likelihood but includes prior information)

LOOCV

Bayesian LOOIC, MCMC version of the algorithm

Algorithm

For observation i in 1:n

- Remove observation i
- Train the model on the remaining n-1 observations
- Derive posterior samples of the predictive density of observation i
- Average over posterior draws
- Take the log

Sum over all n observations

Parallel computation

